

MODULE :- Unit Test Frameworks

EXPERIMENT TITLE :- Data-Driven Automation (DDT) (Assignment 8)

1.Problem Statement :-

To automate login test cases using **Selenium WebDriver and Data-Driven Testing**. The objective is to read multiple sets of username and password combinations from a CSV file, execute the login test for each set of data, and verify whether the login is successful or rejected according to the expected result.

2. Tools and Software Required

S.	Tool/Software	Purpose
1	Python	Programming language used for automation
2	VS Code	IDE for writing and executing the scripts
3	Google Chrome	Browser used for testing
4	Selenium WebDriver	Automates browser interactions
5	ChromeDriver	Connects Selenium with Chrome
6	CSV File	Stores multiple test data sets
7	Python csv module	Reads test data from the CSV file
8	SauceDemo Website	Application used for login testing

3. Concepts Used

- Selenium WebDriver
- **Data-Driven Testing (DDT)**
- CSV file handling
- `csv.DictReader()`
- Test data parameterization
- Selenium locators
- `By.ID`
- `By.NAME`
- `By.XPATH`
- `By.CSS_SELECTOR`
- Assertions
- Positive and negative test cases
- Browser automation
- Test result validation

4.Source Code :-

File 1: test_data.csv

```
username,password,expected_result
standard_user,secret_sauce,success
standard_user,wrong_password,failure
wrong_user,secret_sauce,failure
locked_out_user,secret_sauce,failure
,secret_sauce,failure
standard_user,,failure
```

File 2: test_login.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import csv
import time
```

```
# -----
# Read CSV file
# -----
```

```
def read_test_data():
```

```
    test_data = []
```

```
    with open(
```

```
"test_data.csv",  
"r",  
newline="",  
encoding="utf-8"  
) as file:
```

```
    reader = csv.DictReader(file)
```

```
    for row in reader:  
        test_data.append(row)
```

```
return test_data
```

```
# -----  
# Login Test  
# -----
```

```
def test_login(username, password, expected_result):
```

```
    driver = webdriver.Chrome()  
    driver.maximize_window()
```

```
    try:
```

```
        driver.get(  
            "https://www.saucedemo.com/"  
        )
```

```
        # Username  
        username_field = driver.find_element(  
            By.ID,  
            "user-name"  
        )
```

```
        username_field.send_keys(username)
```

```
        # Password  
        password_field = driver.find_element(  
            By.NAME,  
            "password"  
        )
```

```
        password_field.send_keys(password)
```

```
        # Login  
        login_button = driver.find_element(  
            By.XPATH,  
            "//input[@id='login-button']"  
        )
```

```
        login_button.click()
```

```
        time.sleep(2)
```

```
# -----  
# Validate result
```

```

# -----

if expected_result == "success":

    assert "/inventory.html" in driver.current_url

    print(
        f"PASS: Valid login - {username}"
    )

else:

    error_message = driver.find_element(
        By.CSS_SELECTOR,
        "h3[data-test='error']"
    )

    assert error_message.is_displayed()

    print(
        f"PASS: Invalid login rejected - {username}"
    )

finally:

    driver.quit()

# -----
# Execute all CSV test cases
# -----

if __name__ == "__main__":

    test_cases = read_test_data()

    print(
        "\nTotal test cases:",
        len(test_cases)
    )

    for case in test_cases:

        print("\n-----")
        print(
            "Username:",
            case["username"]
        )
        print(
            "Expected:",
            case["expected_result"]
        )
        print("-----")

        test_login(
            case["username"],
            case["password"],

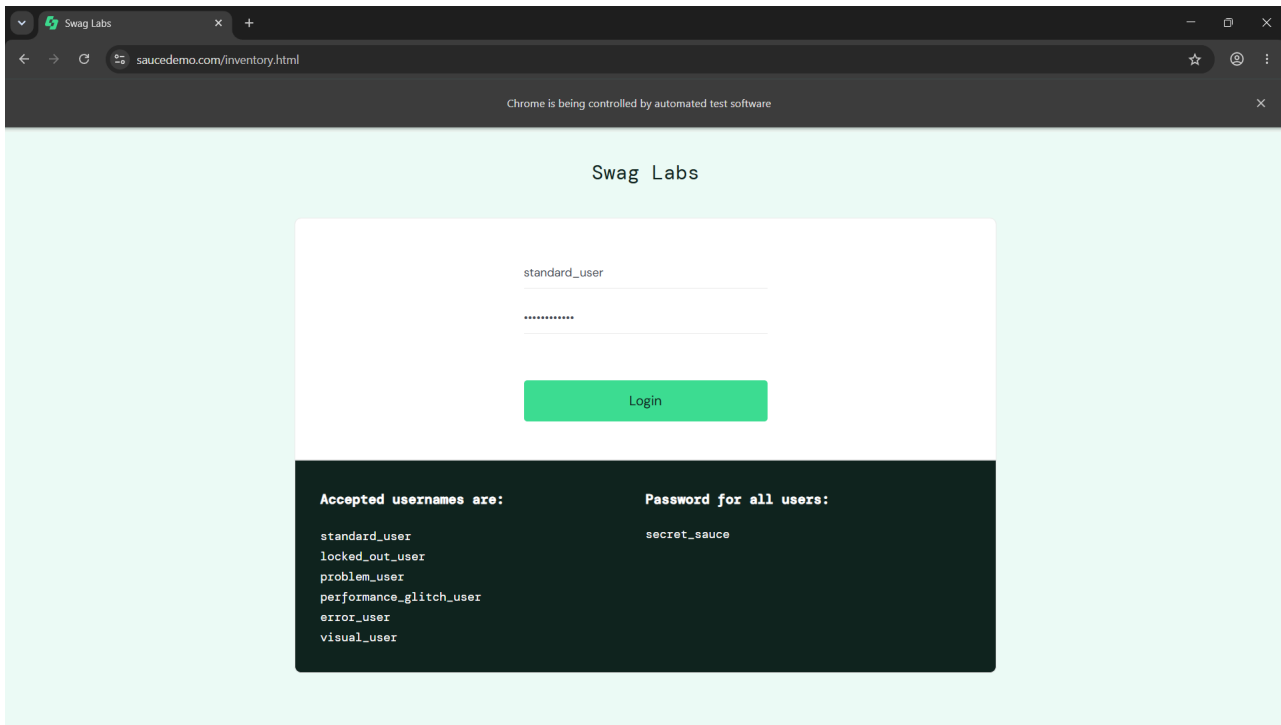
```

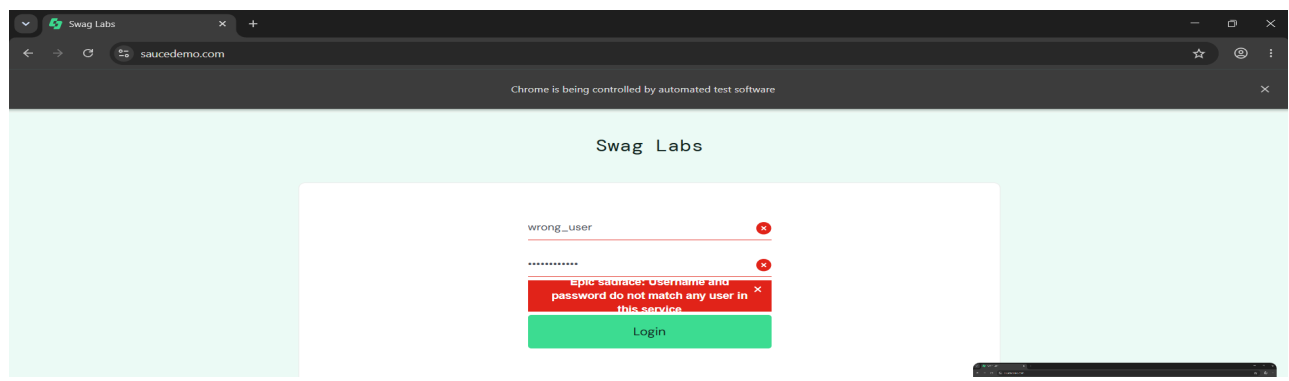
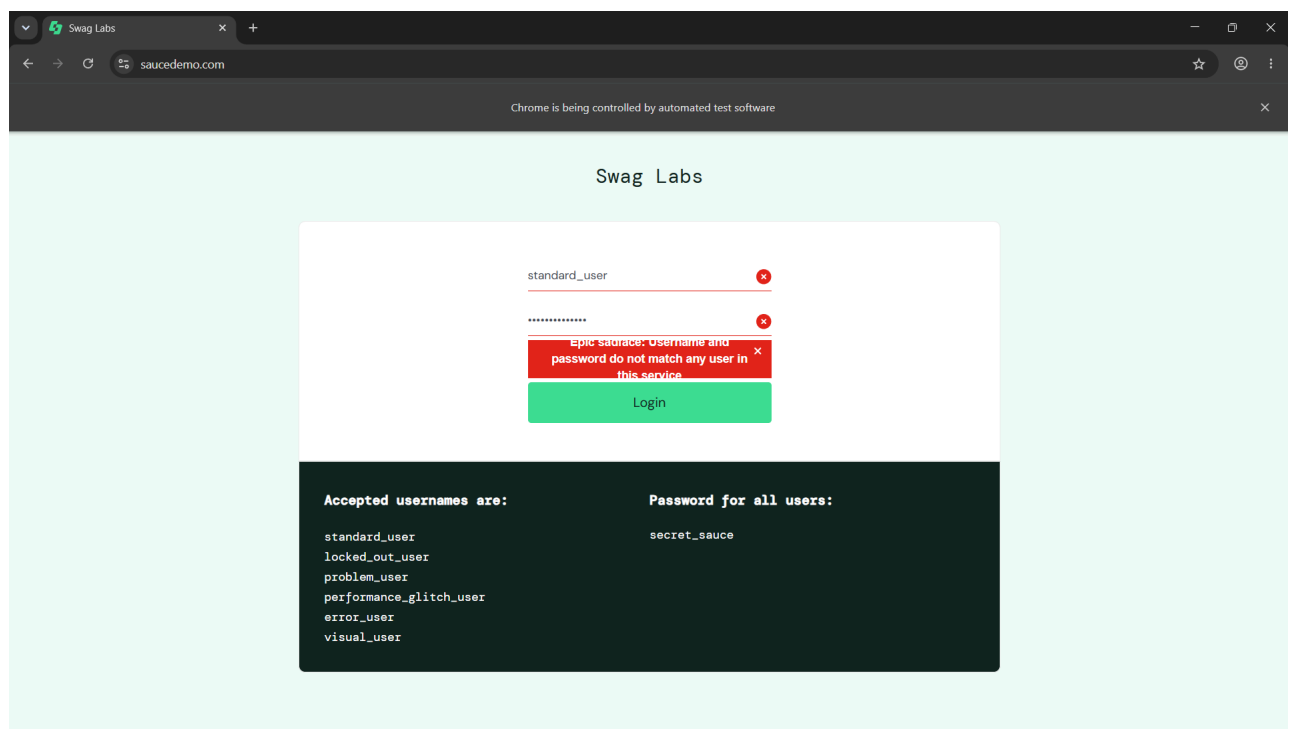
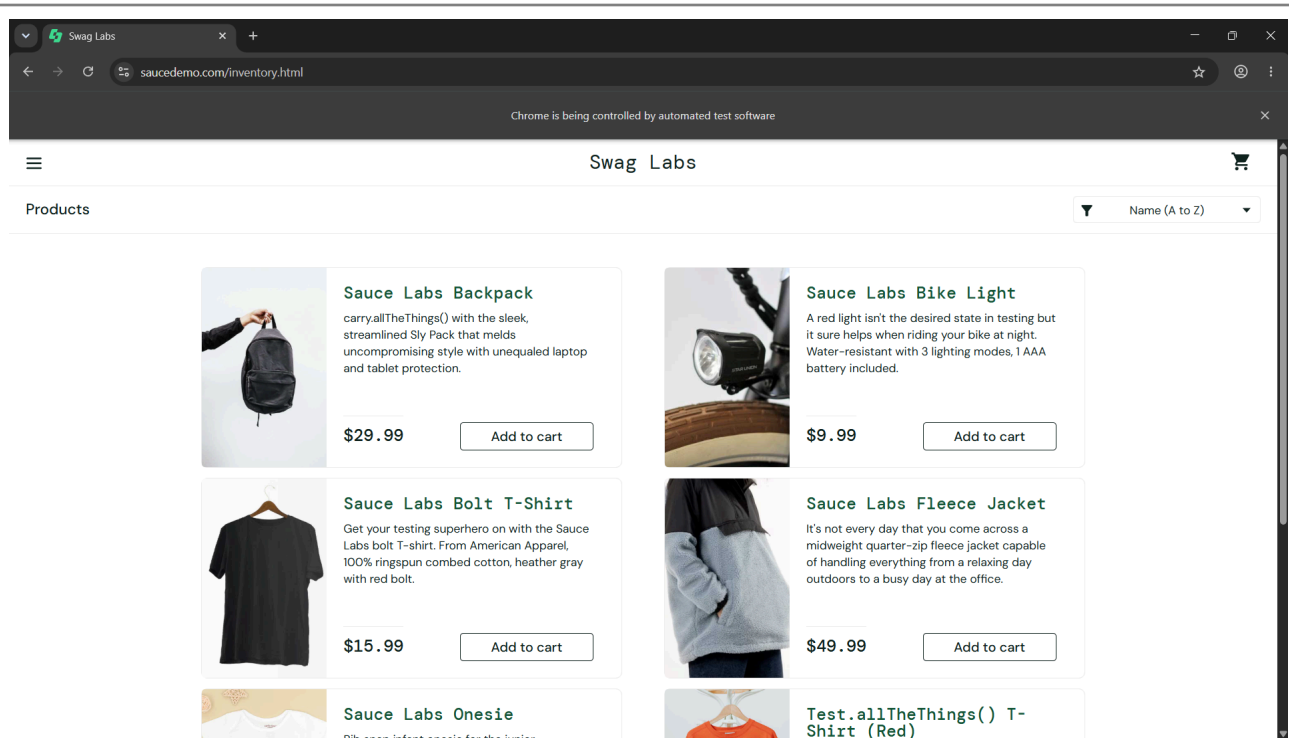
```
case["expected_result"]  
)
```

5.Implementation Steps :-

- 1.Install Python and Selenium WebDriver.
- 2.Create a separate folder for Assignment 8.
- 3.Create a test_data .csv file containing username, password, and expected result.
- 4.Add multiple test cases to the CSV file, including valid and invalid login credentials.
- 5.Create the Python automation script.
- 6.Use Python's `csv.DictReader()` to read the test data from the CSV file.
- 7.Open the SauceDemo login page using Selenium.
- 8.Enter the username and password from each CSV record.
- 9.Click the Login button.
- 10.For a **success** case, verify that the URL contains `/inventory.html`.
- 11.For a **failure** case, verify that an error message is displayed.
- 12.Repeat the test for all test data records.
- 13.Display the result of each test case in the terminal.
- 14.Close the browser after each test execution.
- 15.Verify that all test cases have been executed successfully.

6.Output :-





```
Problems 2 Output Debug Console Terminal Ports powershell - assignment_8 + v [ ] [ ] ... [ ] [X]
```

```
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments> cd assignment_8  
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments\assignment_8> python test_login.py  
  
Total test cases: 6  
  
-----  
Username: standard_user  
Expected: success  
-----  
PASS: Valid login - standard_user  
  
-----  
Username: standard_user  
Expected: failure  
-----  
PASS: Invalid login rejected - standard_user  
  
-----  
Username: wrong_user  
Expected: failure  
-----  
PASS: Invalid login rejected - wrong_user  
  
-----  
Username: locked_out_user  
Expected: failure  
-----  
PASS: Invalid login rejected - locked_out_user  
  
-----  
Username:  
Expected: failure  
-----  
PASS: Invalid login rejected -  
  
-----  
Username: standard_user  
Expected: failure  
-----  
PASS: Invalid login rejected - standard_user  
  
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments\assignment_8>
```